# Supporting Information: Catalytic competence bounds the decay rates of a thermally formed radical pair

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## S1 Software and environment

Spin dynamics: `qbscreen` (this work) with NumPy and SciPy; coherent yields from the Sylvester equation $H_{\mathrm{eff}}X - XH_{\mathrm{eff}}^{\dagger} = -i\rho_0$ with $H_{\mathrm{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P)$ and $X = \int_0^\infty \rho\, dt$, solved by the Bartels–Stewart method;[1] yields with electron spin relaxation from the Liouville-space generator. The two agree to $10^{-8}$ without relaxation, and every yield pair sums to one. Electronic structure: PySCF 2.12.1[2] (B3LYP[3, 4]; def2-SVP for couplings and energies, def2-TZVP[5] for hyperfine couplings; RI-J; PCM), geometries from GFN2-xTB 6.7.1.[6] Every table below is generated from the stored results by `qbscreen/make_si_v6.py`.

## S2 The inequality $\Phi_P \le 4k_P/(k_S + 4k_P)$ and the dead-end bound

Let $\mathcal{L}$ be the generator of eq 2 of the main text, with $\mathcal{D}$ trace preserving, generating a positivity-preserving evolution (for example a trace-preserving GKSL generator[7, 8]) and unital, $\mathcal{D}(I) = 0$.

*Positivity of $e^{\mathcal{L}t}$.* The part $\mathcal{L} - \mathcal{D}$ generates $\rho \mapsto e^{-iH_{\mathrm{eff}}t}\rho\, e^{iH_{\mathrm{eff}}^{\dagger}t}$ with $H_{\mathrm{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P I)$, which is positive; by the Lie–Trotter product formula, $e^{\mathcal{L}t} = \lim_{n\to\infty}(e^{\mathcal{D}t/n}e^{(\mathcal{L}-\mathcal{D})t/n})^n$ is a limit of positive maps and hence positive. For $k_P > 0$ the resolvent $\mathcal{R} = (-\mathcal{L})^{-1} = \int_0^\infty e^{\mathcal{L}t}\, dt$ exists and is positive, because the trace of $e^{\mathcal{L}t}(\rho)$ decays at least as $e^{-k_P t}$ for positive $\rho$, and it maps Hermitian operators to Hermitian operators.

*Two identities.* Unitality gives $-\mathcal{L}(I) = k_S P_S + k_P I$, and applying $\mathcal{R}$,

$$k_S\, \mathcal{R}(P_S) + k_P\, \mathcal{R}(I) = I. \tag{S1}$$

With $\langle A, B\rangle = \mathrm{Tr}[A\,\mathcal{R}(B)]$ (real for Hermitian $A$, $B$), $\Phi_S = (k_S/N)\langle P_S, P_S\rangle$ and $\Phi_P = (k_P/N)\langle I, P_S\rangle$. Trace preservation gives $\Phi_S + \Phi_P = 1$, that is $k_S\langle P_S, P_S\rangle + k_P\langle I, P_S\rangle = N$, while multiplying eq S1 by $P_S$ and taking the trace gives $k_S\langle P_S, P_S\rangle + k_P\langle P_S, I\rangle = N$. Hence $\langle P_S, I\rangle = \langle I, P_S\rangle \equiv c$, and the trace of eq S1 gives $k_S c + k_P\langle I, I\rangle = 4N$.

*Dissipativity.* For Hermitian $B$, $\mathrm{Tr}[B\,\mathcal{L}(B)] = -k_S\mathrm{Tr}(P_S B^2) - k_P\mathrm{Tr}B^2 + \mathrm{Tr}[B\,\mathcal{D}(B)]$, because $\mathrm{Tr}(B[H, B]) = 0$. The map $e^{\mathcal{D}t}$ is positive and unital, so Kadison's inequality[9] gives $(e^{\mathcal{D}t}B)^2 \preceq e^{\mathcal{D}t}(B^2)$; taking the trace and using trace preservation, $f(t) = \mathrm{Tr}(e^{\mathcal{D}t}B)^2 \le \mathrm{Tr}B^2 = f(0)$ for $t \ge 0$, so $f'(0) = 2\mathrm{Tr}[B\,\mathcal{D}(B)] \le 0$. Thus $\mathrm{Tr}[B\,\mathcal{L}(B)] \le 0$. For Hermitian $A$ put $C = \mathcal{R}(A)$; then $\langle A, A\rangle = \mathrm{Tr}[(-\mathcal{L})(C)\,C] \ge k_P\mathrm{Tr}\,C^2 > 0$ for $A \ne 0$. The symmetric part of $\langle\cdot,\cdot\rangle$ is therefore positive definite and satisfies the Cauchy–Schwarz inequality, with equality only for linearly dependent arguments.

*The bound.* Cauchy–Schwarz with the symmetric values $\langle P_S, I\rangle = c$, $\langle P_S, P_S\rangle = (N - k_P c)/k_S$ and $\langle I, I\rangle = (4N - k_S c)/k_P$ gives $k_S k_P c^2 \le (N - k_P c)(4N - k_S c)$, that is $c(k_S + 4k_P) \le 4N$, strictly because $P_S$ and $I$ are linearly independent, and so

$$\Phi_P = \frac{k_P c}{N} \le \frac{4k_P}{k_S + 4k_P}. \tag{S2}$$

Since $\Phi_P = k_P \int \mathrm{Tr}\rho\, dt$ and $\Phi_S = k_S \int \mathrm{Tr}(P_S\rho)\, dt$ with $\Phi_S = 1 - \Phi_P$, the bound is equivalent to $\int \mathrm{Tr}\rho\, dt \le 4\int \mathrm{Tr}(P_S\rho)\, dt$: a singlet-born pair spends at least a quarter of its lifetime in the singlet state.

*Checks.* The function `_selfcheck` of `qbscreen/competence_sensitivity.py` verifies $\langle P_S, I\rangle = \langle I, P_S\rangle$, the dissipativity of the dense Liouvillian and the bound for 20 random spin systems with and without dephasing. `random_bound_test` (seed 20261001; inputs and outputs in `calibration_results/phi_bound_test.json`) evaluates $\Phi_P(k_S+4k_P)/4k_P$ for 300 random spin systems: 0–3 nuclei of spin $\frac{1}{2}$ or 1 with couplings of 1–300 MHz, exchange $10^{-1}$–$10^4$ MHz of either sign, axial dipolar tensors of $-10^{-1}$ to $-10^3$ MHz with random axes, random fields up to 0.1 T and rates, and, for about 30 % of the small systems, single-electron dephasing with $T_2^e$ from 1 ns to 10 $\mu$s. The largest value is 0.999996. `general_H_check` maximizes the same ratio over general Hermitian $H$ (not only spin Hamiltonians) on the two-electron space and with one spin-$\frac{1}{2}$ nucleus, and over $k_S$ and $k_P$, from 60 random starts each. The optimizer approaches one and exceeds it by at most $3 \times 10^{-11}$, which is rounding error (`calibration_results/phi_bound_generalH.json`).

*Sharpness.* For $H = 0$, no nuclei and the fully depolarizing generator below, the populations give $\Phi_P = 4k_P(k_P+\gamma)/[4k_P(k_S+k_P)+\gamma(k_S+4k_P)]$, which equals $k_P/(k_S+k_P)$ at $\gamma = 0$ and approaches $4k_P/(k_S+4k_P)$ as $\gamma \to \infty$ without reaching it. With $k_S = 1$ and $k_P = 10^{-6}$ $\mu$s$^{-1}$, the fully depolarizing generator $\mathcal{D}(\rho) = \gamma(\mathrm{Tr}\rho\, I/4-\rho)$, which is trace preserving and unital, gives $\Phi_P(k_S+4k_P)/4k_P = 0.25$ at $\gamma = 0$ and 0.999999 at $\gamma = 10^4$ $\mu$s$^{-1}$ (`competence_sensitivity.sharpness_example`; `calibration_results/phi_bound_sharpness.json`).

*Trace preservation is needed.* The unital, completely positive but population-creating generator $\mathcal{D}(\rho) = \gamma(V\rho V^\dagger - \frac{1}{2}\{VV^\dagger, \rho\})$ with $V = |T_0\rangle\langle S|$ gives $\Phi_P = 150/101$ at $k_S = 0.01$, $k_P = \gamma = 1$ $\mu$s$^{-1}$ (`competence_sensitivity.trace_counterexample`), which violates both $\Phi_P \le 1$ and the bound above.

For a dead-end pair ($k_P = 0$) the resolvent need not exist: without nuclei and with $H = 0$, for example, the $16 \times 16$ generator has rank 7, because triplet states that never reach the singlet are stationary. Integrating $\frac{d}{du}e^{\mathcal{L}_0 u}(I) = -k_S\, e^{\mathcal{L}_0 u}(P_S)$ instead gives

$$k_S \int_0^t e^{\mathcal{L}_0 u}(P_S)\, du = I - e^{\mathcal{L}_0 t}(I) \preceq I \tag{S3}$$

for all $t$, since $e^{\mathcal{L}_0 t}(I) \succeq 0$. The population built up from a constant singlet source $k_f[\mathrm{E}\cdot\mathrm{S}]P_S/N$ is therefore at most $4k_f[\mathrm{E}\cdot\mathrm{S}]/k_S = e^{-\Delta/k_BT}[\mathrm{E}\cdot\mathrm{S}]$ at all times.

## S3 The search with the exchange coupling tied to $k_S$

The model has flavin N5 and N10 with the isotropic couplings of ref. [10] plus the density-functional spin-dipolar tensors of Table S9, rotated into the frame of the contact complex, and the two aminium $\alpha$-protons (isotropic). For each $(k_S, k_P)$ on a grid of 15 by 13 rates ($10^{-2}$–$10^5$ and $10^{-2}$–$10^4$ $\mu$s$^{-1}$, half-decade steps), the exchange coupling is scanned on 99 points with sinh spacing (40 MHz scale), symmetric about zero and extending to $\max(4000\ \mathrm{MHz}, 5(k_S+k_P)/2\pi)$, and zoomed twice on 21-point subgrids around each of the three largest local maxima. At every $J$ the response is maximized over $s = 0.2$, 0.35, 0.6 and 1 and 25 field directions (the pole and eight azimuths over $2\pi$ at polar angles of 30, 60 and 90°; the yield is invariant under $B \to -B$), and the maximum, its $s$ and direction, and the zero-field onward yield are stored (`qbscreen/constrained_map.py`; `calibration_results/constrained_map.json`). For a pair at free energy $\Delta$ and a set of reorganization energies, the admitted exchange couplings form the interval spanned by eq 7 of the main text over those $\lambda$ for which $k_S \le 0.3\,\nu_n e^{-(\lambda-\Delta_S)^2/4\lambda k_BT}$, $\nu_n = 10^{13}$ s$^{-1}$; $F$ is the largest of the stored values inside that interval and of the values at its ends, interpolated linearly in the stored $F(J)$; a fixed $\lambda$ admits a single $J$, and couplings beyond the scanned range take the edge value. The general curves are evaluated at 310 K, the measured MAO B rates at 298.15 K, the temperature of the measurements. `constrained_map.interp_check` evaluates the response directly at the admitted $J$, over the same four $s$ values and 25 directions, for 40 cells and driving forces drawn at random within 0.05 eV of the fixed-$\lambda$ thresholds for $k_{\mathrm{cat}} = 1$ s$^{-1}$; the direct values are $2 \times 10^{-4}$ to 1.21 times the interpolated ones, and multiplying every fixed-$\lambda$ value by 1.21 moves the fixed-$\lambda$ thresholds by at most 0.02 eV (`constrained_map.interp_shift`; `calibration_results/constrained_interp_check.json`). Because the admitted set depends on $\Delta$ through eq 7, $G$ need not decrease with $\Delta$; we report its running maximum from larger $\Delta$, which leaves every threshold

unchanged and, for $\lambda = 0.2$–$2.0$ eV and for $J$ free, every value. `constrained_map.s_profile` gives $F_{\mathrm{free}}$ at the threshold-setting cell as a function of $s$ (Table S3; `calibration_results/constrained_s_profile.json`). At $\Delta = 0.5$ eV, 0 of the 195 cells admit no $\lambda$ in 0.2–2.0 eV. Table S1 lists $F_{\mathrm{free}}$, the maximum over all scanned $J$, and Table S2 the thresholds of $G$ for several sets of $\lambda$ and rates (`constrained_map.analyze`; `calibration_results/constrained_G.json`). At the 0.1 % threshold for $k_{\mathrm{cat}} = 1\ \mathrm{s}^{-1}$, $G$ is set by the cell $k_S = 316$, $k_P = 31.6\ \mu\mathrm{s}^{-1}$, whose zero-field onward yield is 0.19; the absolute change of $\Phi_P$ there is $1 \times 10^{-4}$.

Table S1: $F_{\mathrm{free}}(k_S, k_P)$ (%) of the search of section S3 ($J$ unconstrained, $s \in [0.2, 1]$). Rates in $\mu\mathrm{s}^{-1}$.

| $\log k_P \backslash \log k_S$ | -2.0 | -1.5 | -1.0 | -0.5 | 0.0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -2.0 | 1.3 | 2.1 | 2.2 | 2.1 | 1.8 | 1.1 | 0.76 | 0.6 | 0.54 | 0.52 | 0.47 | 0.45 | 0.49 | 0.44 | 0.44 |
| -1.5 | 0.41 | 1.3 | 2.1 | 2.2 | 1.9 | 1.5 | 0.95 | 0.75 | 0.6 | 0.54 | 0.52 | 0.47 | 0.48 | 0.44 | 0.44 |
| -1.0 | 0.14 | 0.36 | 0.74 | 2 | 1.6 | 1.5 | 1.3 | 0.94 | 0.75 | 0.59 | 0.52 | 0.49 | 0.46 | 0.44 | 0.4 |
| -0.5 | 0.045 | 0.13 | 0.36 | 0.74 | 1.3 | 1.5 | 1.4 | 1.2 | 0.94 | 0.75 | 0.54 | 0.46 | 0.45 | 0.4 | 0.24 |
| 0.0 | 0.014 | 0.044 | 0.13 | 0.34 | 0.7 | 1.2 | 1.1 | 1.2 | 1 | 0.82 | 0.63 | 0.49 | 0.39 | 0.23 | 0.091 |
| 0.5 | 0.0036 | 0.011 | 0.035 | 0.1 | 0.26 | 0.44 | 0.56 | 0.69 | 0.72 | 0.56 | 0.43 | 0.36 | 0.2 | 0.082 | 0.028 |
| 1.0 | 3.8e-4 | 0.0012 | 0.0037 | 0.011 | 0.032 | 0.075 | 0.14 | 0.2 | 0.23 | 0.23 | 0.17 | 0.12 | 0.048 | 0.025 | 0.015 |
| 1.5 | 1.7e-5 | 5.4e-5 | 1.7e-4 | 5.3e-4 | 0.0016 | 0.0046 | 0.012 | 0.029 | 0.051 | 0.06 | 0.046 | 0.029 | 0.018 | 0.0084 | 0.0034 |
| 2.0 | 1.1e-6 | 3.4e-6 | 1.1e-5 | 3.4e-5 | 1.1e-4 | 3.3e-4 | 1.0e-3 | 0.0028 | 0.0067 | 0.011 | 0.011 | 0.0077 | 0.0045 | 0.0017 | 3.9e-4 |
| 2.5 | 6.2e-8 | 1.9e-7 | 6.2e-7 | 1.9e-6 | 6.1e-6 | 1.9e-5 | 6.0e-5 | 1.8e-4 | 5.1e-4 | 0.0011 | 0.0018 | 0.002 | 0.0012 | 3.2e-4 | 5.1e-5 |
| 3.0 | 3.8e-9 | 1.2e-8 | 3.8e-8 | 1.2e-7 | 3.8e-7 | 1.2e-6 | 3.8e-6 | 1.2e-5 | 3.6e-5 | 9.5e-5 | 1.9e-4 | 2.1e-4 | 1.0e-4 | 2.4e-5 | 3.3e-6 |
| 3.5 | 1.0e-10 | 3.2e-10 | 1.0e-9 | 3.2e-9 | 1.0e-8 | 3.2e-8 | 1.0e-7 | 3.1e-7 | 9.4e-7 | 2.6e-6 | 5.3e-6 | 5.3e-6 | 2.2e-6 | 2.6e-7 | 4.9e-8 |
| 4.0 | 9.5e-13 | 2.4e-12 | 7.3e-12 | 2.3e-11 | 7.2e-11 | 2.3e-10 | 7.2e-10 | 2.3e-9 | 7.0e-9 | 2.1e-8 | 5.7e-8 | 1.1e-7 | 1.1e-7 | 3.2e-8 | 3.0e-9 |

Table S2: Driving force (eV) at which $G$ falls below 0.1 % and 0.01 %, for the admitted reorganization energies (or $J$ unconstrained) and for the rate $k$ the pair must carry. "–": not reached within the range where $G$ is defined.

| rate carried | 0.2–2.0 eV | 0.5 eV | 1.0 eV | 1.5 eV | $J$ free |
|---|---|---|---|---|---|
| 1 $\mathrm{s}^{-1}$ | 0.47 / 0.52 | 0.46 / 0.50 | 0.47 / 0.49 | 0.41 / 0.46 | 0.47 / 0.52 |
| 10 $\mathrm{s}^{-1}$ | 0.40 / 0.46 | 0.40 / 0.46 | 0.40 / 0.43 | 0.33 / 0.37 | 0.40 / 0.46 |
| 100 $\mathrm{s}^{-1}$ | 0.34 / 0.40 | 0.34 / 0.40 | 0.33 / 0.37 | 0.26 / 0.27 | 0.34 / 0.40 |
| 1000 $\mathrm{s}^{-1}$ | 0.28 / 0.33 | 0.28 / 0.33 | 0.26 / 0.29 | 0.15 / 0.18 | 0.28 / 0.33 |
| 12.7 $\mathrm{s}^{-1}$ (benzylamine, 298 K) | 0.38 / 0.43 | 0.38 / 0.43 | 0.38 / 0.40 | 0.30 / 0.32 | 0.38 / 0.43 |
| 572 $\mathrm{s}^{-1}$ (phenylethylamine, 298 K) | 0.29 / 0.34 | 0.29 / 0.34 | 0.26 / 0.29 | 0.15 / 0.19 | 0.29 / 0.34 |

Table S3: $F_{\mathrm{free}}$ at $k_S = 316$, $k_P = 31.6\ \mu\mathrm{s}^{-1}$, the cell that sets the 0.1 % threshold for $k_{\mathrm{cat}} = 1\ \mathrm{s}^{-1}$, as a function of the dipolar scale $s$ (same $J$ scan and directions; one $s$ value per search).

| $s$ | $F_{\mathrm{free}}$ (%) |
|---|---|
| 0 | 0.19 |
| 0.01 | 0.21 |
| 0.03 | 0.17 |
| 0.1 | 0.085 |
| 0.2 | 0.06 |
| 0.35 | 0.039 |
| 0.6 | 0.025 |
| 1 | 0.017 |
| 2 | 0.0063 |
| 3 | 0.0049 |

## S4 Search with $J$ free and $s \in [0, 3]$

As a comparison, the model with isotropic flavin couplings was searched with $J$ unconstrained and the dipolar scale over $s \in [0, 3]$. For each $(k_S, k_P)$ on the same grid, the exchange coupling is scanned on 99 points with sinh spacing (40 MHz scale), symmetric about zero and extending to $\max(4000\ \mathrm{MHz}, 5(k_S + k_P)/2\pi)$, for dipolar scales $s = 0$, 0.3, 1 and 3 and 25 field directions (the pole and eight azimuths

over $2\pi$ at polar angles of 30, 60 and 90°; the yield is invariant under $B \to -B$). Each of the three largest local maxima in $J$ is refined twice on 21-point subgrids, and the four best distinct candidates are polished by a Nelder–Mead search in $(J, \theta, \phi, s)$ with $s$ clipped to $[0, 3]$ (160 evaluations each). A cell whose maximum lies within 5 % of the range limit is searched again with a range of sixteen times the position of the maximum. Because the response has narrow resonances in the dipolar scale (at $k_S = 10^4$, $k_P = 10^{-2}$ $\mu s^{-1}$ and $J \approx 0$ it is 0.40, 0.14, 0.68 and 0.04 % at $s = 0$, 0.003, 0.01 and 0.03 at the first-pass optimum; `calibration_results/s_resonance_example.json`), every cell is then searched a second time: $s$ on 41 values (0 and 40 logarithmically spaced from $10^{-3}$ to 3), $J$ on 21 points of the same sinh grid plus the first-pass optimum, all 25 directions, followed by a polish of the four best candidates (`worst_case_map.supplement`). The cell keeps the larger value. The second pass raised 151 cells, by up to a factor of 3.6, without changing any row maximum by more than 0.0 %. Repeating it with 121 values of $s$ at the cells that set the thresholds changed $F$ by a factor of at most 1.018 (`calibration_results/s_convergence.json`). Table S4 lists $F$ in percent.

With these values and the competence condition of the main text, the 0.1 % threshold for $k_{\mathrm{cat}} = 1\ \mathrm{s}^{-1}$ is 0.47 eV.

Table S4: $F(k_S, k_P)$ (%) of the search of section S4 ($J$ free, $s \in [0, 3]$, isotropic flavin couplings). Rates in $\mu s^{-1}$.

| $\log k_P \backslash \log k_S$ | -2.0 | -1.5 | -1.0 | -0.5 | 0.0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -2.0 | 6.3 | 12 | 14 | 13 | 10 | 7.5 | 6 | 5.2 | 4.6 | 3.7 | 2.3 | 1.3 | 1.3 | 1.3 | 1.3 |
| -1.5 | 2.5 | 6.2 | 11 | 14 | 13 | 9.6 | 7.1 | 5.3 | 3.8 | 2.3 | 1.3 | 1.3 | 1.3 | 1.2 | 1.2 |
| -1.0 | 0.85 | 2.4 | 6 | 11 | 13 | 12 | 8.4 | 5.4 | 2.7 | 1.1 | 1 | 1.1 | 1.1 | 1 | 1 |
| -0.5 | 0.26 | 0.8 | 2.3 | 5.5 | 9.7 | 11 | 9 | 5.2 | 2 | 0.83 | 0.9 | 0.78 | 0.63 | 0.54 | 0.42 |
| 0.0 | 0.069 | 0.22 | 0.66 | 1.9 | 4.4 | 7.2 | 7.4 | 4.4 | 1.6 | 0.59 | 0.62 | 0.63 | 0.59 | 0.43 | 0.31 |
| 0.5 | 0.014 | 0.044 | 0.14 | 0.41 | 1.1 | 2.5 | 3.6 | 2.7 | 1.1 | 0.37 | 0.34 | 0.37 | 0.35 | 0.26 | 0.15 |
| 1.0 | 0.0014 | 0.0044 | 0.014 | 0.043 | 0.13 | 0.34 | 0.66 | 0.75 | 0.48 | 0.2 | 0.07 | 0.084 | 0.088 | 0.077 | 0.045 |
| 1.5 | 5.8e-5 | 1.8e-4 | 5.8e-4 | 0.0018 | 0.0056 | 0.017 | 0.043 | 0.083 | 0.096 | 0.062 | 0.032 | 0.049 | 0.055 | 0.042 | 0.02 |
| 2.0 | 1.7e-6 | 5.4e-6 | 1.7e-5 | 5.4e-5 | 1.7e-4 | 5.3e-4 | 0.0015 | 0.004 | 0.0077 | 0.014 | 0.018 | 0.02 | 0.023 | 0.015 | 0.0043 |
| 2.5 | 1.2e-7 | 3.8e-7 | 1.2e-6 | 3.8e-6 | 1.2e-5 | 3.7e-5 | 1.2e-4 | 3.5e-4 | 9.6e-4 | 0.0021 | 0.0033 | 0.0039 | 0.0036 | 0.0014 | 2.6e-4 |
| 3.0 | 4.8e-9 | 1.5e-8 | 4.8e-8 | 1.5e-7 | 4.8e-7 | 1.5e-6 | 4.8e-6 | 1.5e-5 | 4.4e-5 | 1.1e-4 | 2.3e-4 | 2.6e-4 | 1.4e-4 | 3.3e-5 | 4.5e-6 |
| 3.5 | 1.0e-10 | 3.2e-10 | 1.0e-9 | 3.2e-9 | 1.0e-8 | 3.2e-8 | 1.0e-7 | 3.1e-7 | 9.5e-7 | 2.6e-6 | 5.3e-6 | 5.4e-6 | 2.2e-6 | 3.0e-7 | 5.7e-8 |
| 4.0 | 1.1e-12 | 2.6e-12 | 7.6e-12 | 2.3e-11 | 7.4e-11 | 2.3e-10 | 7.4e-10 | 2.3e-9 | 7.2e-9 | 2.2e-8 | 5.8e-8 | 1.2e-7 | 1.1e-7 | 3.3e-8 | 3.2e-9 |

## S5 Dependence on the spin model

In the anisotropic model "aniso" described below, the full search of section S4 (first pass, edge rule and fine-$s$ pass) was repeated on every cell with $k_P = 10$–$10^{2.5}$ $\mu s^{-1}$ and $k_S \geq k_P$ (`model_sensitivity.run_aniso_full`; `calibration_results/worst_case_v6_aniso.json`); Table S5 lists the values, 0 of which lie at the edge of the $J$ range. The thresholds in the main text replace each of these cells by the larger of its reference and anisotropic values.

The other models were examined only locally: for each onward rate that sets the thresholds of $G$, the cell that gives the row maximum of $F$ in the reference model was searched again (`qbscreen/model_sensitivity.py`), starting from the reference optimum and a 39-point $J$ grid at its field direction and dipolar scale, and polishes the two best candidates. In "aniso" N5 and N10 carry the literature isotropic couplings plus the density-functional spin-dipolar tensors (Table S9), rotated into the frame of the contact complex in which the dipolar tensor was computed (0.09 Å root-mean-square deviation between the relaxed anion and the complex flavin). The complex was built with methylamine, so the benzylaminium tensors have no orientation in it: the aminium $\alpha$-protons are kept isotropic (their anisotropy is below 5 % of the isotropic part), and in "+$^{14}$N" the aminium nitrogen (spin 1), whose tensor is nearly axial about the singly occupied orbital, is added with its axis along the lone pair of the placed amine. "+2H" adds the two largest flavin methyl protons (Table S6). Because only the reference row-maximum cell was searched, and only near its optimum, another cell of the same row may give a larger value; the "aniso" column is the local reference for the two richer models, and the full anisotropic search is given above.

Table S5: $F(k_S, k_P)$ (%) in the anisotropic model from the full search. Rates in $\mu s^{-1}$.

| $\log k_P \backslash \log k_S$ | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|---|---|---|
| 1.0 | 1.7 | 2.3 | 1.8 | 0.99 | 0.52 | 0.28 | 0.18 | 0.14 | 0.067 |
| 1.5 | | 0.23 | 0.28 | 0.21 | 0.13 | 0.068 | 0.026 | 0.017 | 0.0082 |
| 2.0 | | | 0.022 | 0.027 | 0.021 | 0.011 | 0.0049 | 0.0019 | 5.1e-4 |
| 2.5 | | | | 0.0021 | 0.003 | 0.0025 | 0.0014 | 3.7e-4 | 6.4e-5 |

Table S6: Searched $F$ (%) at the threshold-setting cells in the reference and richer models (local searches).

| $\log k_P$ | $\log k_S$ | reference | aniso | $+^{14}$N | +2H |
|---|---|---|---|---|---|
| 1.0 | 1.5 | 0.75 | 2.3 | 1 | 1.6 |
| 1.5 | 2.0 | 0.096 | 0.28 | 0.16 | 0.2 |
| 2.0 | 4.0 | 0.023 | 0.0035 | 0.0033 | 0.0036 |
| 2.5 | 3.5 | 0.0039 | 0.0023 | 0.0015 | 0.0022 |

## S6 Electron spin relaxation in the four-nucleus model

The four-nucleus spin space has dimension 144, too large for the dense Liouville-space solution. With $\gamma = 2/T_2^e$, single-electron dephasing $\gamma \sum_e (S_{ez}\rho S_{ez} - \rho/4)$ is split into a decay $-(\gamma/2)\rho$, absorbed into $K = 2\pi H - \frac{i}{2}[k_S P_S + (k_P + \gamma/2)I]$, and a recycling term $\gamma \sum_e S_{ez}\rho S_{ez}$. The integral $X = \int_0^\infty \rho\, dt$ then solves $KX - XK^\dagger = -i\rho_0 - i\gamma \sum_e S_{ez} X S_{ez}$, which is iterated by GMRES[11] with the left side inverted by a triangular Sylvester solve in the Schur basis of $K$ (`product_yield.yields_t2e`). On twelve random systems small enough for the dense solution, including anisotropic hyperfine tensors, the two agree to a relative $1.3 \times 10^{-13}$. For every cell with $k_S \geq k_P$ on the threshold rows, the first-pass search of section S4 was repeated with $T_2^e = 10$ ns, 100 ns and 1 $\mu$s, with the same edge rule for the $J$ range (Table S7; `calibration_results/worst_case_v6_t2e.json`). Nineteen searches (14 at $T_2^e = 10$ ns, 5 at 100 ns) first peaked at the edge of the $J$ range and were repeated with the range widened sixteenfold; one of them ($k_S = 10^{4.5}$, $k_P = 10$ $\mu s^{-1}$, $T_2^e = 100$ ns) still peaks at the edge because the response approaches a plateau as $|J| \to \infty$: it is constant to five figures for $|J| = 10^6$–$10^8$ MHz and equal to the stored value (`calibration_results/t2e_plateau_check.json`).

## S7 Hyperfine couplings

Fermi-contact couplings (B3LYP/def2-TZVP at GFN2-xTB geometries), all nuclei with $|a_{\mathrm{iso}}| \geq 1$ MHz (Table S8), and the spin-dipolar tensors of the nitrogens and the largest protons (Table S9), computed from the spin density with the operator-derivative integrals $\langle \mu | \partial_a \partial_b |\mathbf{r} - \mathbf{R}|^{-1} | \nu \rangle$; their trace reproduces $-4\pi\rho_s(\mathbf{R})$. The computed flavin N5 and N10 isotropic couplings are within 16 % and 13 % of the literature values used in the main text,[10] and the N5 tensor is nearly axial with its large component normal to the ring. For 3-methylindole$^{\bullet+}$ the two large methyl couplings stand in for the tryptophan $\beta$-protons; their values depend on the side-chain conformation.

## S8 Dipolar tensor

The tensor is the point-dipole tensor summed over pairs of atoms weighted by their Löwdin[12] spin populations (UKS B3LYP/def2-SVP on each radical at its position in the complex), for methylamine on the flavin si face with N5–N = 3.5 Å (Table S10).

Table S7: Searched $F$ (%) without and with single-electron dephasing in the four-nucleus model. Rates in $\mu s^{-1}$.

| $\log k_P$ | $\log k_S$ | coherent | 10 ns | 100 ns | 1000 ns |
|---|---|---|---|---|---|
| 1.0 | 1.0 | 0.66 | 0.075 | 0.22 | 0.54 |
| 1.0 | 1.5 | 0.75 | 0.12 | 0.29 | 0.63 |
| 1.0 | 2.0 | 0.48 | 0.18 | 0.26 | 0.41 |
| 1.0 | 2.5 | 0.2 | 0.21 | 0.18 | 0.18 |
| 1.0 | 3.0 | 0.07 | 0.22 | 0.21 | 0.067 |
| 1.0 | 3.5 | 0.084 | 0.22 | 0.21 | 0.073 |
| 1.0 | 4.0 | 0.088 | 0.22 | 0.21 | 0.076 |
| 1.0 | 4.5 | 0.077 | 0.22 | 0.21 | 0.065 |
| 1.0 | 5.0 | 0.045 | 0.22 | 0.21 | 0.04 |
| 1.5 | 1.5 | 0.083 | 0.024 | 0.046 | 0.077 |
| 1.5 | 2.0 | 0.096 | 0.042 | 0.059 | 0.09 |
| 1.5 | 2.5 | 0.062 | 0.063 | 0.045 | 0.059 |
| 1.5 | 3.0 | 0.032 | 0.072 | 0.028 | 0.03 |
| 1.5 | 3.5 | 0.049 | 0.075 | 0.035 | 0.047 |
| 1.5 | 4.0 | 0.055 | 0.076 | 0.04 | 0.053 |
| 1.5 | 4.5 | 0.042 | 0.077 | 0.033 | 0.041 |
| 1.5 | 5.0 | 0.02 | 0.077 | 0.015 | 0.019 |
| 2.0 | 2.0 | 0.0077 | 0.0067 | 0.0068 | 0.0076 |
| 2.0 | 2.5 | 0.014 | 0.011 | 0.01 | 0.014 |
| 2.0 | 3.0 | 0.018 | 0.013 | 0.014 | 0.017 |
| 2.0 | 3.5 | 0.02 | 0.015 | 0.016 | 0.019 |
| 2.0 | 4.0 | 0.023 | 0.015 | 0.018 | 0.022 |
| 2.0 | 4.5 | 0.015 | 0.015 | 0.012 | 0.014 |
| 2.0 | 5.0 | 0.0043 | 0.015 | 0.0033 | 0.0042 |
| 2.5 | 2.5 | 0.0021 | 0.0016 | 0.002 | 0.0021 |
| 2.5 | 3.0 | 0.0033 | 0.0024 | 0.0031 | 0.0032 |
| 2.5 | 3.5 | 0.0039 | 0.0023 | 0.0036 | 0.0039 |
| 2.5 | 4.0 | 0.0036 | 0.0017 | 0.0031 | 0.0035 |
| 2.5 | 4.5 | 0.0014 | 0.0013 | 0.0012 | 0.0013 |
| 2.5 | 5.0 | 2.6e-4 | 0.0012 | 2.2e-4 | 2.5e-4 |

## S9 Electronic coupling

Fragment-orbital couplings[13] $t = \langle \mathrm{LUMO_{Fl}} | F | \mathrm{HOMO_{am}} \rangle$ (B3LYP/def2-SVP) for methylamine and benzylamine stacked on the flavin si face (five orientations) or approaching its edge (three orientations), at N5–N distances of 3.0–6.0 Å. Geometries with any intermolecular contact below 2.0 Å are excluded (Table S11). The edge approach of benzylamine clashes at every distance below 6 Å and is not used.

## S10 Density-functional driving force

The main text uses measured potentials. For comparison, Table S12 gives adiabatic one-electron potentials in water ($\varepsilon = 78.4$, absolute SHE potential 4.44 V[14]) from the same density-functional protocol, against the measured values for a dialkylamine[14] and for free FMN at pH 7.[15] The flavin comparison is between different species (lumiflavin anion radical versus the pH-7 FMN oxidized/semiquinone couple) and indicates only the order of the error. Table S13 gives the raw driving force in low-dielectric continua with a contact Coulomb term $-e^2/(4\pi\varepsilon_0\varepsilon r)$. No zero-point or thermal corrections are included, and none of these values enters the main text.

## S11 The driving force in the active site

In eq 10 of the main text only $E^\circ_{\mathrm{aq}}$ of the amine and the enzyme-bound flavin potential are measured. In scenario (a) the Coulomb term in water is $-e^2/(4\pi\varepsilon_0 \cdot 78.4\, r)$, $r = 3.5$–6 Å. In scenario (c) the site

Table S8: Isotropic hyperfine couplings.

| radical | nucleus (atom index) | $a_{\mathrm{iso}}$ (MHz) | $a_{\mathrm{iso}}$ ($\mu$T) |
|---|---|---|---|
| lumiflavin$^{\bullet-}$ | H29 | +17.9 | +639 |
| | H30 | +17.8 | +636 |
| | N4 | +12.4 | +441 |
| | H25 | +11.3 | +403 |
| | H26 | +11.3 | +402 |
| | H22 | -10.5 | -374 |
| | H20 | -5.3 | -190 |
| | H21 | -5.3 | -190 |
| | N13 | +4.6 | +165 |
| | H27 | +1.4 | +50 |
| | N8 | -1.0 | -36 |
| methylaminium$^{\bullet+}$ | H2 | +363.1 | +12955 |
| | H6 | +82.7 | +2951 |
| | H5 | +81.9 | +2923 |
| | H3 | -59.3 | -2116 |
| | H4 | -59.3 | -2115 |
| | N1 | +30.5 | +1090 |
| benzylaminium$^{\bullet+}$ | H10 | +256.9 | +9167 |
| | H11 | +196.2 | +7001 |
| | N0 | +24.8 | +885 |
| | H8 | -23.0 | -821 |
| | H9 | -20.7 | -737 |
| | H14 | -12.3 | -439 |
| | H16 | -7.8 | -278 |
| | H13 | -3.2 | -115 |
| | H15 | +2.4 | +85 |
| 3-methylindole$^{\bullet+}$ | H12 | +60.1 | +2143 |
| | H11 | +55.5 | +1980 |
| | H18 | -14.3 | -511 |
| | H13 | -13.1 | -467 |
| | H14 | -13.1 | -466 |
| | H16 | -9.8 | -351 |
| | H15 | -4.9 | -175 |
| | N3 | +4.8 | +172 |
| | H17 | +2.7 | +95 |

has an effective dielectric constant $\varepsilon = 4$ or 10. The aminium radical cation is desolvated with a Born term $e^2/(8\pi\varepsilon_0 a)\,(1/\varepsilon - 1/78.4)$, where the radius $a = 2.81$ Å reproduces, by the Born formula in water, the solvation-energy difference of 244 kJ mol$^{-1}$ between dimethylamine and its radical cation,[14] and the Coulomb term changes by $-e^2/(4\pi\varepsilon_0 r)\,(1/\varepsilon - 1/78.4)$. No correction is applied to the flavin, whose potential is measured in the enzyme. The change of $\Delta$ is $-0.37$ to $+0.04$ eV over $\varepsilon$ and $r$; it is negative because the stronger attraction of the pair outweighs the desolvation of the aminium. These continuum values are illustrative; we have not shown that they bracket the true transfer free energies. Moving the flavin anion into the same continuum as well, with the density-functional protocol of the previous section, would raise $\Delta$ by $+0.01$ to $+0.56$ eV, but it would count the protein environment of the flavin, already contained in its measured potential, a second time, and we do not use it.

*Substrate-induced shift.* Without substrate the oxidized/semiquinone and semiquinone/reduced midpoints of the MAO A flavin are $-159$ and $-262$ mV,[16] so the two-electron midpoint is $-211$ mV. With substrate the full-reduction midpoint shifts by up to $+0.5$ V and no semiquinone is observed.[16] If the absence of semiquinone reflects equilibrium, the first one-electron potential lies at or below the two-electron midpoint, at most $-211 + 500$ mV $= +0.29$ V, a shift of the one-electron couple of at most 0.45 V. The main text uses the full $+0.5$ V, which lowers $\Delta$ more than this reading allows. All three estimates are computed in `qbscreen/make_numbers_v6.py`.

Table S9: Spin-dipolar hyperfine tensors.

| radical | nucleus | $a_{\mathrm{iso}}$ (MHz) | principal values of $\mathsf{A}_{\mathrm{dip}}$ (MHz) |
|---|---|---|---|
| lumiflavin$^{\bullet-}$ | H29 | +17.9 | -1.2, -1.0, +2.2 |
| | H30 | +17.8 | -1.2, -1.0, +2.2 |
| | N4 | +12.4 | -17.0, -16.7, +33.7 |
| | H25 | +11.3 | -1.5, -0.9, +2.4 |
| | H26 | +11.3 | -1.5, -0.9, +2.4 |
| | N13 | +4.6 | -6.0, -5.7, +11.7 |
| | N8 | -1.0 | -0.2, +0.1, +0.1 |
| | N11 | -0.2 | -0.6, -0.4, +1.1 |
| benzylaminium$^{\bullet+}$ | H10 | +256.9 | -8.0, -4.2, +12.2 |
| | H11 | +196.2 | -7.9, -4.2, +12.1 |
| | N0 | +24.8 | -18.8, -18.6, +37.4 |
| | H8 | -23.0 | -21.8, -6.0, +27.8 |
| | H9 | -20.7 | -21.4, -5.4, +26.8 |

Table S10: Dipolar coupling of the contact pair.

| model | $D$ (MHz) | $E$ (MHz) | principal values of $\mathsf{T}$ (MHz) |
|---|---|---|---|
| spin-population tensor | -843 | -26 | -1124, 511, 613 |
| point dipole at spin centroids | -1642 | 0 | – |
| point dipole at N5 and N | -1817 | 0 | – |

## S12 Cryptochrome

Orientation-averaged, perpendicular and parallel response of the cryptochrome pair over the exchange prefactor $J_0$ (both signs of $J_{\mathrm{EH}}$; $J = -2J_{\mathrm{EH}}$, $J_{\mathrm{EH}} = J_0 e^{-\beta r}$, $\beta = 14$ nm$^{-1}$, $r = 1.90$ nm[17]), $k_S = k_P = 1$ $\mu$s$^{-1}$, coherent (Table S14).

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Table S11: Coupling scan. "valid": non-clashing geometries of those computed.

| amine | approach | $r_{\mathrm{N5N}}$ (Å) | valid | median $\lvert t\rvert$ (meV) | max $\lvert t\rvert$ (meV) | min contact (Å) |
|---|---|---|---|---|---|---|
| methylamine | face | 3.0 | 5/5 | 184.6 | 311.6 | 2.32 |
| methylamine | face | 3.5 | 5/5 | 52.8 | 105.6 | 2.72 |
| methylamine | face | 4.0 | 5/5 | 24.6 | 37.9 | 3.12 |
| methylamine | face | 4.5 | 5/5 | 5.1 | 31.1 | 3.53 |
| methylamine | face | 5.0 | 5/5 | 1.9 | 14.7 | 3.96 |
| methylamine | face | 5.5 | 5/5 | 1.0 | 4.2 | 4.40 |
| methylamine | face | 6.0 | 5/5 | 0.3 | 1.1 | 4.85 |
| methylamine | edge | 3.0 | 0/3 | – | – | 0.77 |
| methylamine | edge | 3.5 | 0/3 | – | – | 0.93 |
| methylamine | edge | 4.0 | 0/3 | – | – | 1.29 |
| methylamine | edge | 4.5 | 2/3 | 97.0 | 170.8 | 1.71 |
| methylamine | edge | 5.0 | 3/3 | 23.1 | 81.0 | 2.17 |
| methylamine | edge | 5.5 | 3/3 | 8.6 | 35.0 | 2.65 |
| methylamine | edge | 6.0 | 3/3 | 3.2 | 12.7 | 3.13 |
| benzylamine | face | 3.0 | 0/5 | – | – | 0.61 |
| benzylamine | face | 3.5 | 2/5 | 47.1 | 52.3 | 0.89 |
| benzylamine | face | 4.0 | 2/5 | 5.2 | 7.9 | 1.31 |
| benzylamine | face | 4.5 | 3/5 | 6.1 | 35.2 | 1.77 |
| benzylamine | face | 5.0 | 5/5 | 4.2 | 21.1 | 2.25 |
| benzylamine | face | 5.5 | 5/5 | 1.0 | 8.3 | 2.73 |
| benzylamine | face | 6.0 | 5/5 | 0.9 | 2.0 | 3.22 |
| benzylamine | edge | 3.0 | 0/3 | – | – | 1.00 |
| benzylamine | edge | 3.5 | 0/3 | – | – | 1.00 |
| benzylamine | edge | 4.0 | 0/3 | – | – | 0.92 |
| benzylamine | edge | 4.5 | 0/3 | – | – | 0.82 |
| benzylamine | edge | 5.0 | 0/3 | – | – | 1.00 |
| benzylamine | edge | 5.5 | 0/3 | – | – | 1.36 |
| benzylamine | edge | 6.0 | 1/3 | 0.4 | 0.4 | 1.78 |

Table S12: One-electron potentials in water.

| couple | DFT (V vs NHE) | experiment | error (V) |
|---|---|---|---|
| $Me_2NH^{\bullet+}/Me_2NH$ | +0.73 | +1.300 | -0.57 |
| $MeNH_2^{\bullet+}/MeNH_2$ | +1.26 | – | – |
| $BnNH_2^{\bullet+}/BnNH_2$ | +1.45 | – | – |
| $Fl/Fl^{\bullet-}$ (exp.: FMN, pH 7) | -1.16 | -0.313 | -0.84 |


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Table S13: Raw DFT driving force of the contact pair.

| continuum | amine | separated (eV) | $r = 4$ Å | 5 Å | 6 Å |
|---|---|---|---|---|---|
| $\varepsilon = 4$ | methylamine | 3.55 | 2.65 | 2.83 | 2.95 |
| $\varepsilon = 4$ | benzylamine | 3.59 | 2.69 | 2.87 | 2.99 |
| $\varepsilon = 10$ | methylamine | 2.83 | 2.47 | 2.55 | 2.59 |
| $\varepsilon = 10$ | benzylamine | 2.97 | 2.61 | 2.69 | 2.73 |


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Table S14: Cryptochrome response (%).

| $J_0$ ($\mu$T) | $J$ (MHz) | powder (%) | $\perp$ (%) | $\parallel$ (%) |
|---|---|---|---|---|
| 8.00e12 | -1.26 | +0.417 | +0.494 | +0.027 |
| 8.00e12 | +1.26 | +0.386 | +0.464 | -0.023 |
| 1.01e13 | -1.58 | +0.395 | +0.469 | +0.015 |
| 1.01e13 | +1.58 | +0.384 | +0.457 | -0.030 |
| 1.27e13 | -1.99 | +0.371 | +0.454 | -0.005 |
| 1.27e13 | +1.99 | +0.392 | +0.461 | -0.034 |
| 1.60e13 | -2.51 | +0.403 | +0.498 | +0.005 |
| 1.60e13 | +2.51 | +0.413 | +0.491 | -0.029 |
| 2.01e13 | -3.16 | +0.470 | +0.561 | +0.021 |
| 2.01e13 | +3.16 | +0.435 | +0.536 | -0.020 |
| 2.53e13 | -3.98 | +0.573 | +0.681 | +0.027 |
| 2.53e13 | +3.98 | +0.444 | +0.580 | -0.012 |
| 3.18e13 | -5.01 | +0.634 | +0.751 | +0.042 |
| 3.18e13 | +5.01 | +0.404 | +0.483 | +0.007 |
| 4.01e13 | -6.30 | +0.559 | +0.622 | +0.034 |
| 4.01e13 | +6.30 | +0.334 | +0.407 | +0.025 |
| 5.05e13 | -7.93 | +0.383 | +0.420 | +0.061 |
| 5.05e13 | +7.93 | +0.242 | +0.320 | -0.005 |
| 6.35e13 | -9.99 | +0.441 | +0.534 | +0.033 |
| 6.35e13 | +9.99 | +0.242 | +0.299 | +0.049 |
| 8.00e13 | -12.57 | +0.492 | +0.589 | +0.015 |
| 8.00e13 | +12.57 | +0.168 | +0.234 | +0.021 |
| 1.01e14 | -15.83 | +0.282 | +0.372 | +0.024 |
| 1.01e14 | +15.83 | +0.153 | +0.174 | +0.081 |
| 1.27e14 | -19.93 | +0.085 | +0.115 | -0.060 |
| 1.27e14 | +19.93 | +0.150 | +0.131 | +0.126 |
| 1.60e14 | -25.09 | +0.061 | +0.078 | +0.020 |
| 1.60e14 | +25.09 | +0.185 | +0.119 | +0.214 |
| 2.01e14 | -31.58 | +0.044 | +0.049 | +0.023 |
| 2.01e14 | +31.58 | +0.055 | +0.078 | +0.063 |
| 2.53e14 | -39.76 | -0.009 | -0.004 | -0.013 |
| 2.53e14 | +39.76 | +0.036 | +0.041 | +0.019 |
| 3.18e14 | -50.05 | +0.015 | +0.017 | +0.009 |
| 3.18e14 | +50.05 | +0.068 | +0.063 | +0.041 |
| 4.01e14 | -63.01 | +0.010 | +0.011 | +0.007 |
| 4.01e14 | +63.01 | +0.052 | +0.045 | +0.031 |
| 5.05e14 | -79.33 | +0.004 | +0.005 | +0.003 |
| 5.05e14 | +79.33 | +0.039 | +0.049 | +0.015 |
| 6.35e14 | -99.87 | +0.003 | +0.003 | +0.002 |
| 6.35e14 | +99.87 | +0.034 | +0.036 | +0.019 |
| 8.00e14 | -125.73 | +0.001 | +0.002 | +0.001 |
| 8.00e14 | +125.73 | +0.011 | +0.017 | +0.002 |